

Synoptic CB: Macrophyte Fluxes

2024 Samples

2026-03-16

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#Read in and collate all the files in the Processed Data folder

##Pull in Hydrago probe data

Some flux IDs are missing from hydrago data

```
## [1] "GCW_WC_1_202405" "GCW_WC_1_202409" "GCW_WC_1_202410" "GCW_WC_1_202411"
## [5] "GCW_WC_2_202405" "GCW_WC_2_202409" "GCW_WC_2_202410" "GCW_WC_2_202411"
## [9] "GCW_WC_3_202405" "GCW_WC_3_202409" "GCW_WC_3_202410" "GCW_WC_3_202411"
## [13] "GCW_WC_4_202405" "GCW_WC_4_202409" "GCW_WC_4_202410" "GCW_WC_4_202411"
## [17] "GCW_WC_5_202405" "GCW_WC_5_202409" "GCW_WC_5_202410" "GCW_WC_5_202411"
## [21] "GWI_TR_1_202404" "GWI_TR_1_202409" "GWI_TR_1_202410" "GWI_TR_2_202404"
## [25] "GWI_TR_2_202409" "GWI_TR_2_202410" "GWI_TR_3_202404" "GWI_TR_3_202409"
## [29] "GWI_TR_3_202410" "GWI_TR_4_202404" "GWI_TR_4_202409" "GWI_TR_4_202410"
## [33] "GWI_TR_5_202404" "GWI_TR_5_202409" "GWI_TR_5_202410" "GWI_WC_1_202404"
## [37] "GWI_WC_1_202409" "GWI_WC_1_202410" "GWI_WC_2_202404" "GWI_WC_2_202409"
## [41] "GWI_WC_2_202410" "GWI_WC_3_202404" "GWI_WC_3_202409" "GWI_WC_3_202410"
## [45] "GWI_WC_4_202404" "GWI_WC_4_202409" "GWI_WC_4_202410" "GWI_WC_5_202404"
## [49] "GWI_WC_5_202409" "GWI_WC_5_202410" "MSM_TR_1_202404" "MSM_TR_1_202408"
## [53] "MSM_TR_1_202409" "MSM_TR_1_202410" "MSM_TR_1_202411" "MSM_TR_2_202404"
## [57] "MSM_TR_2_202408" "MSM_TR_2_202409" "MSM_TR_2_202410" "MSM_TR_2_202411"
## [61] "MSM_TR_3_202404" "MSM_TR_3_202408" "MSM_TR_3_202409" "MSM_TR_3_202410"
## [65] "MSM_TR_3_202411" "MSM_TR_4_202404" "MSM_TR_4_202408" "MSM_TR_4_202409"
## [69] "MSM_TR_4_202410" "MSM_TR_4_202411" "MSM_TR_5_202404" "MSM_TR_5_202408"
## [73] "MSM_TR_5_202409" "MSM_TR_5_202410" "MSM_TR_5_202411" "MSM_WC_1_202404"
## [77] "MSM_WC_1_202408" "MSM_WC_1_202409" "MSM_WC_1_202410" "MSM_WC_2_202404"
## [81] "MSM_WC_2_202408" "MSM_WC_2_202409" "MSM_WC_2_202410" "MSM_WC_3_202404"
## [85] "MSM_WC_3_202408" "MSM_WC_3_202409" "MSM_WC_3_202410" "MSM_WC_4_202404"
## [89] "MSM_WC_4_202408" "MSM_WC_4_202409" "MSM_WC_4_202410" "MSM_WC_5_202404"
## [93] "MSM_WC_5_202408" "MSM_WC_5_202409" "MSM_WC_5_202410" "SWH_WC_1_202404"
## [97] "SWH_WC_1_202409" "SWH_WC_1_202410" "SWH_WC_1_202411" "SWH_WC_2_202404"
```

```
## [101] "SWH_WC_2_202409" "SWH_WC_2_202410" "SWH_WC_2_202411" "SWH_WC_3_202404"  
## [105] "SWH_WC_3_202409" "SWH_WC_3_202410" "SWH_WC_3_202411" "SWH_WC_4_202404"  
## [109] "SWH_WC_4_202409" "SWH_WC_4_202410" "SWH_WC_4_202411" "SWH_WC_5_202404"  
## [113] "SWH_WC_5_202409" "SWH_WC_5_202410" "SWH_WC_5_202411"
```

```
## Remove bad fluxes and calculate how many there were
```

```
## Number flux measurements removed: 2
```

```
## Number negative CO2 fluxes removed: 2
```

```
## Number messy fluxes removed: 1
```

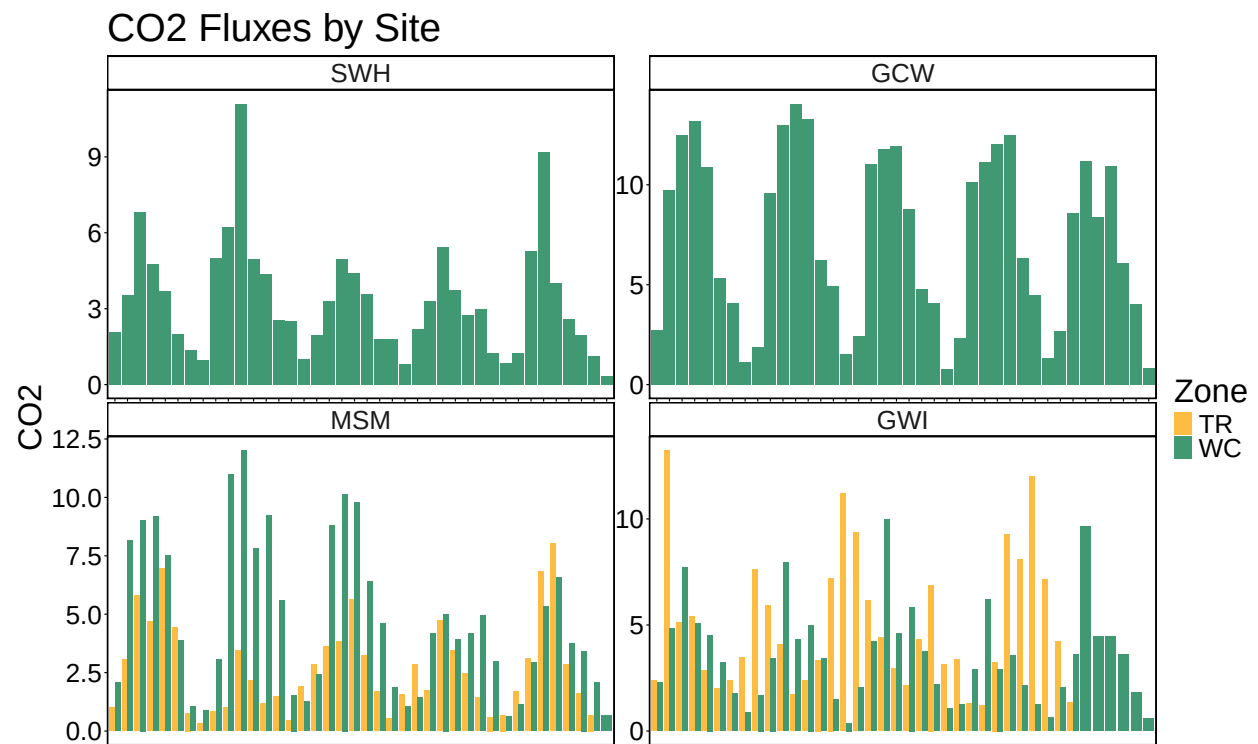
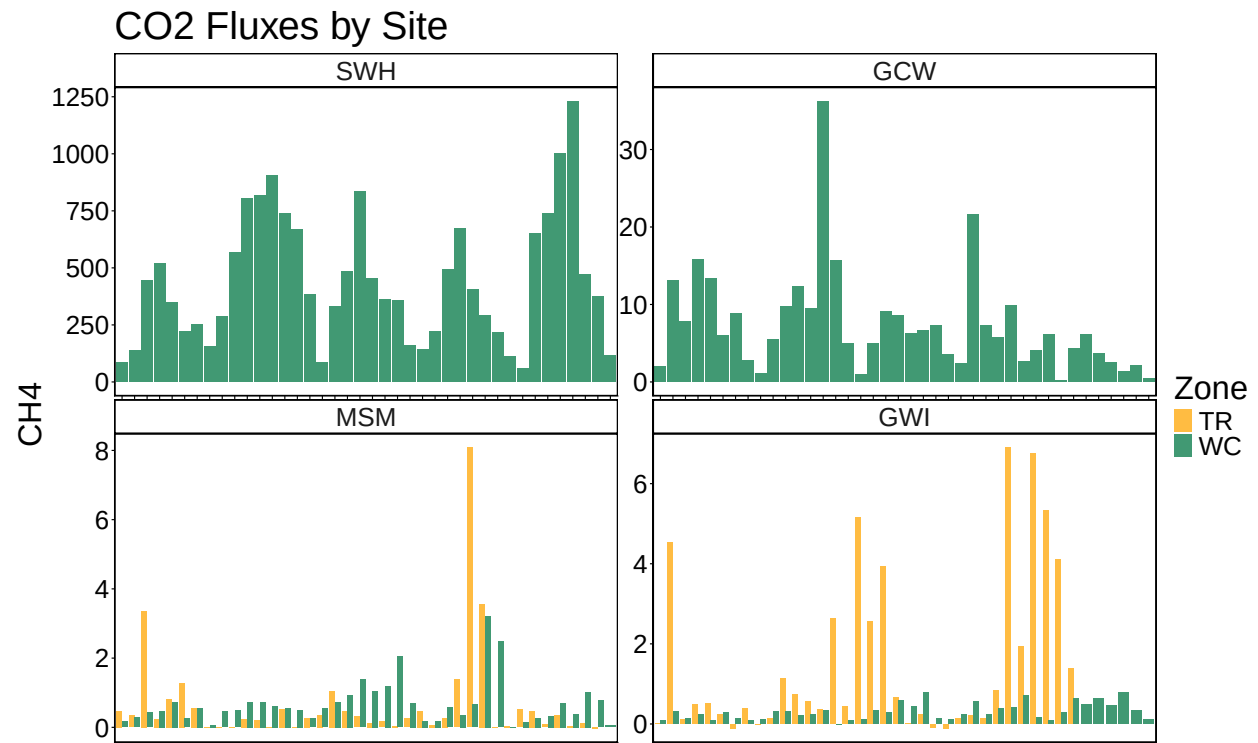
```
## Percentage flux measurements removed: 0.85%
```

```
## Percentage negative CO2 fluxes removed: 0.85%
```

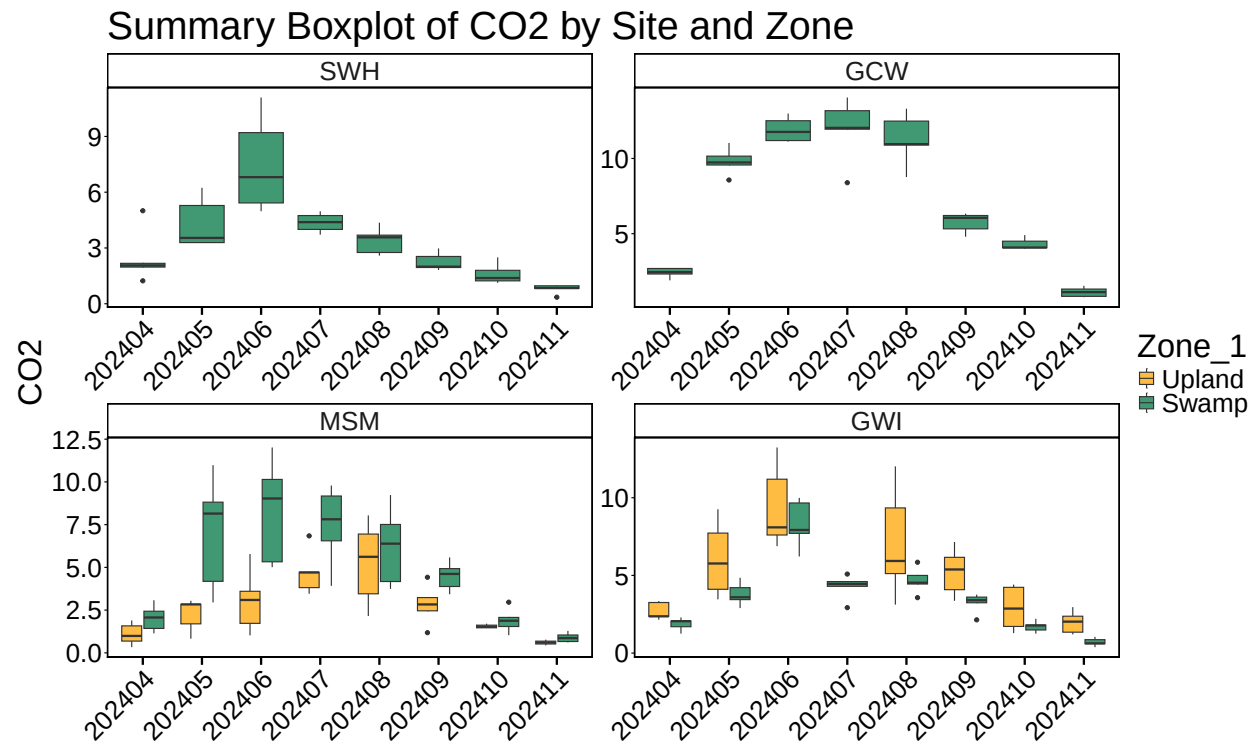
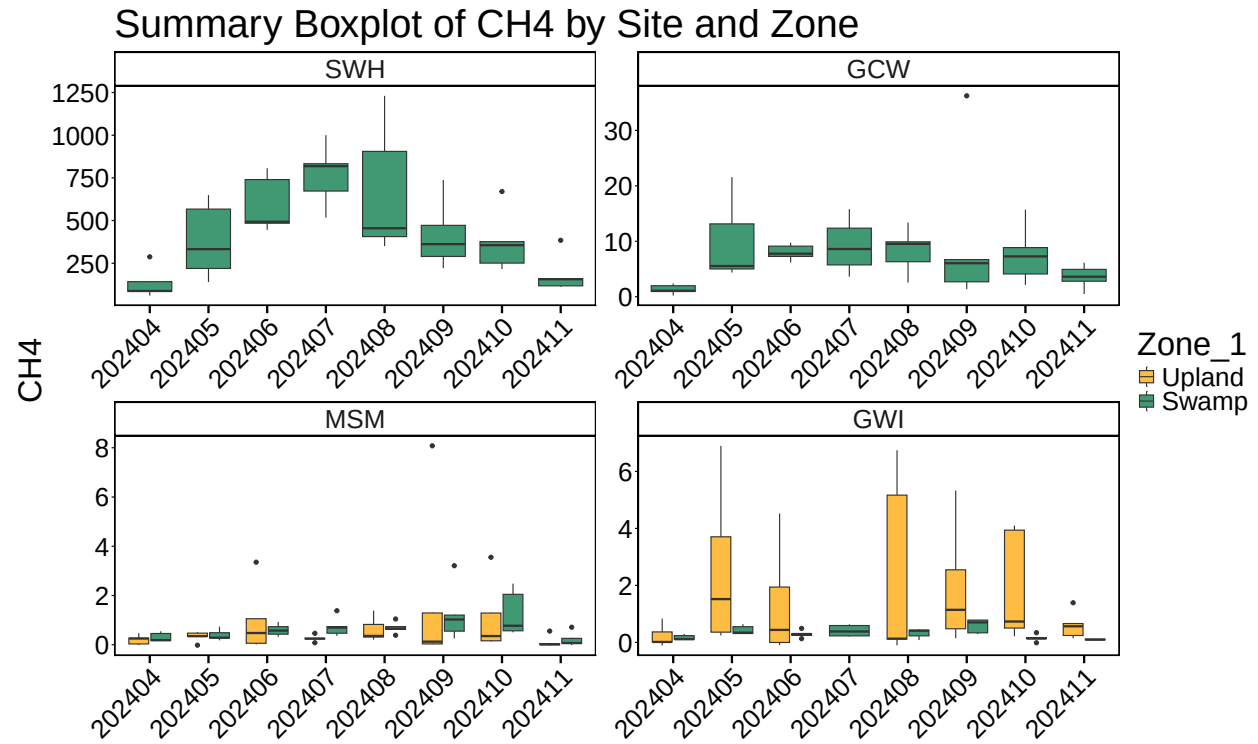
```
## Percentage messy fluxes removed: 0.43%
```

0.1 Write out files

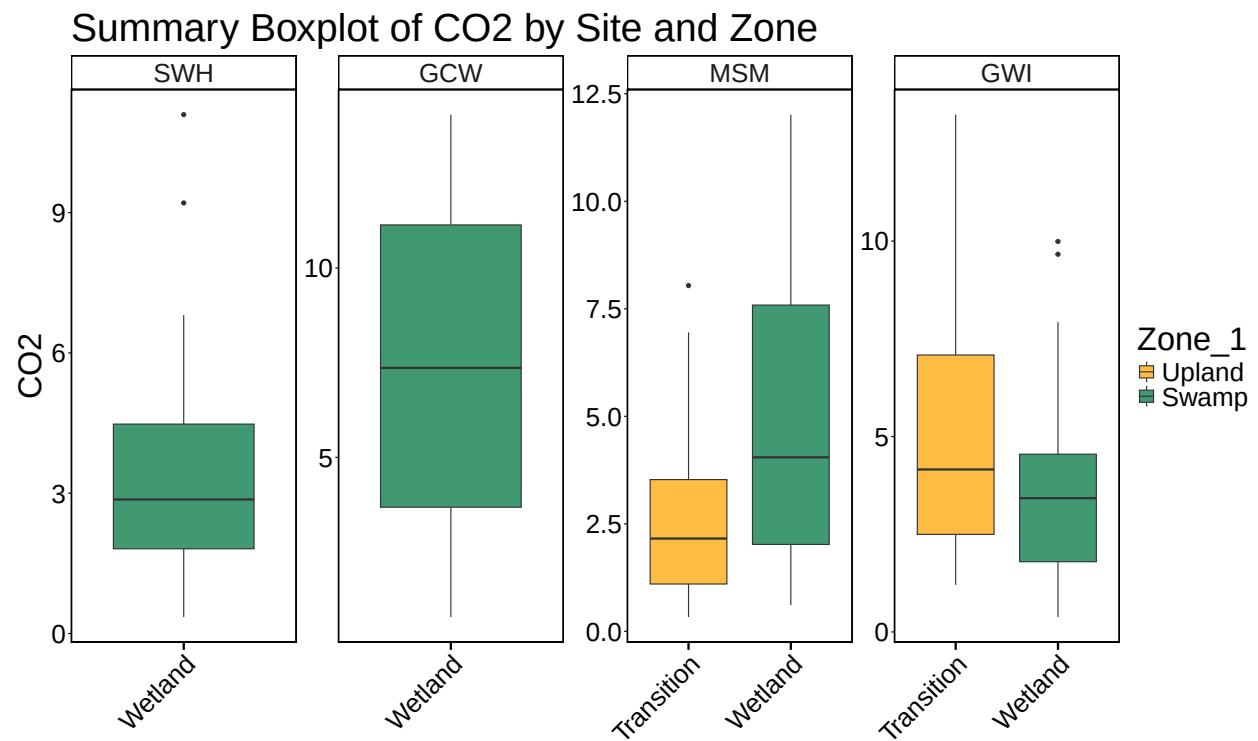
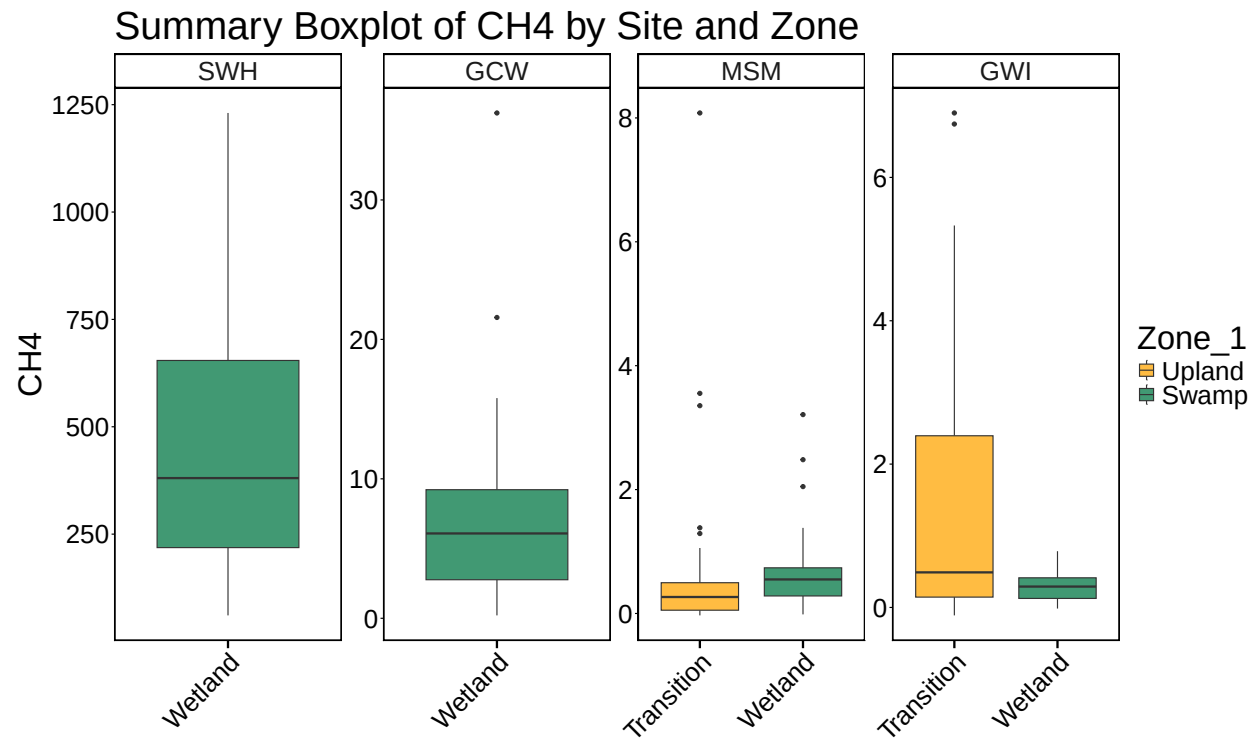
0.2 All Data by Year, Site and Zone



0.3 Boxplot summary by Date



0.4 Boxplot summary by Site



##Calculate summary metrics

0.5 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
SWH	Wetland	60.8085475	1230.676594	440.3399282	0.3521706	11.094771	3.342001
GCW	Wetland	0.2071297	36.236250	7.3249032	0.7823220	14.050978	7.314527
MSM	Transition	-0.0310129	8.079763	0.6801219	0.3322192	8.040166	2.694096
MSM	Wetland	-0.0149613	3.210500	0.6819524	0.6116895	12.015151	4.786112
GWI	Transition	-0.1108124	6.898915	1.5376979	1.2049340	13.234715	5.024564
GWI	Wetland	-0.0154060	0.784701	0.3055206	0.3810429	9.988793	3.570166